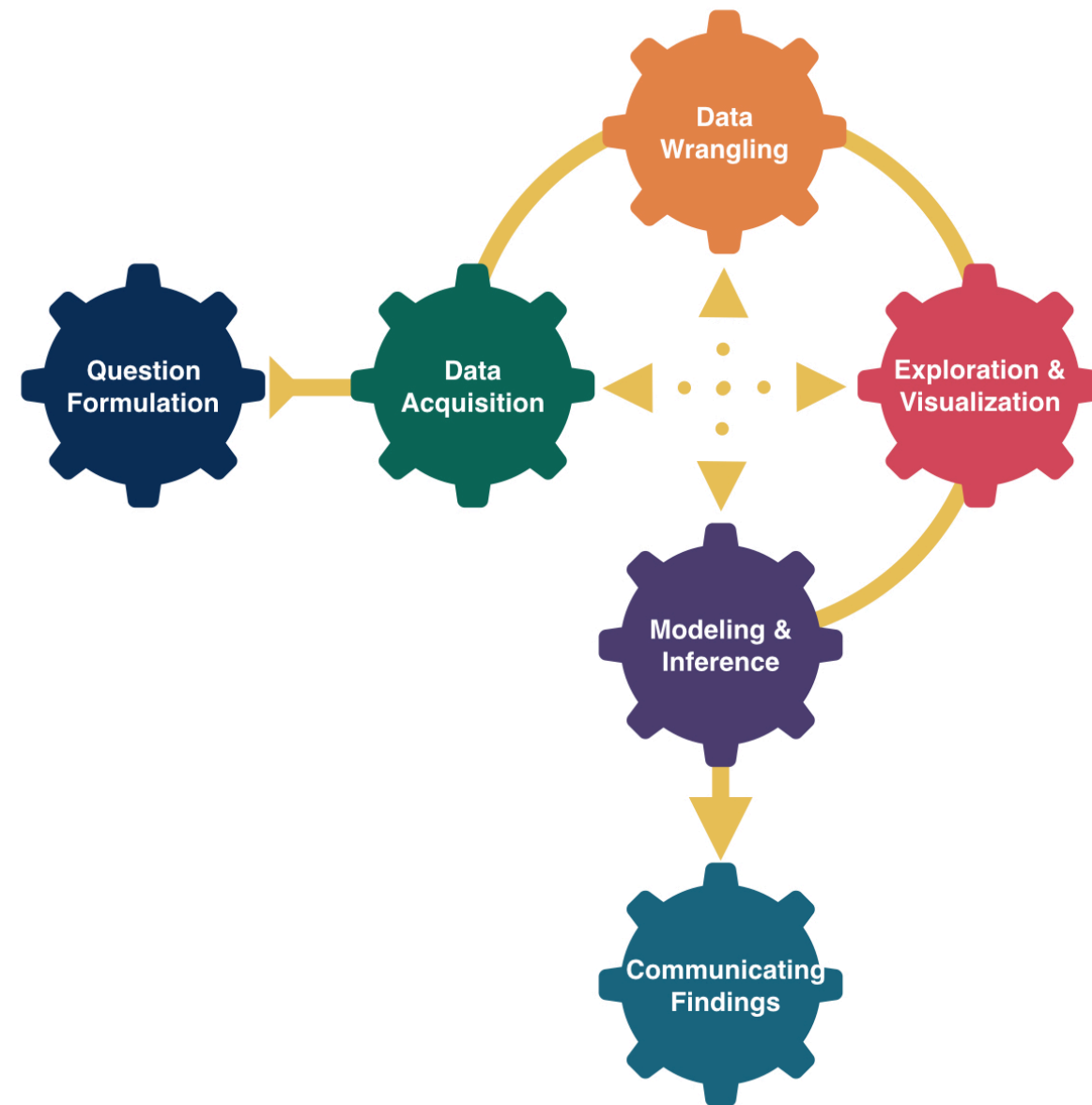


More Data Wrangling

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DATA 100

Week 4 | Spring 2026



Announcements

Code a Valentine: Tomorrow, 4:30pm, Taylor 212

Goals for Today

More data wrangling

Data joins

Load Necessary Packages



dplyr is part of this collection of data science packages.

```
1 # Load necessary packages
2 library(tidyverse)
```

Data Setting: Bureau of Labor Statistics (BLS) Consumer Expenditure Survey

BLS Mission: “Measures labor market activity, working conditions, price changes, and productivity in the U.S. economy to support public and private decision making.”

Data: Last quarter of the 2016 BLS Consumer Expenditure Survey.

```
1 library(tidyverse)
2
3 ce_raw <- read_csv("data/fmli.csv",
4                   na = c("NA", "."))
5 glimpse(ce_raw)
```

Rows: 6,301

Columns: 51

```
$ NEWID      <chr> "03324174", "03324204", "03324214", "03324244", "03324274", "...
$ PRINEARN   <chr> "01", "01", "01", "01", "02", "01", "01", "01", "02", "01", "...
$ FINLWT21   <dbl> 25984.767, 6581.018, 20208.499, 18078.372, 20111.619, 19907.3...
$ FINCBTAX   <dbl> 116920, 200, 117000, 0, 2000, 942, 0, 91000, 95000, 40037, 10...
$ BLS_URBN   <dbl> 1, 1, 1, 1, 1, 1, 1, 1, 2, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1...
$ POPSIZE    <dbl> 2, 3, 4, 2, 2, 2, 1, 2, 5, 2, 3, 2, 2, 3, 4, 3, 3, 1, 4, 1, 1...
$ EDUC_REF   <chr> "16", "15", "16", "15", "14", "11", "10", "13", "12", "12", "...
$ EDUCA2     <dbl> 15, 15, 13, NA, NA, NA, NA, 15, 15, 14, 12, 12, NA, NA, NA, 1...
$ AGE_REF    <dbl> 63, 50, 47, 37, 51, 63, 77, 37, 51, 64, 26, 59, 81, 51, 67, 4...
$ AGE2       <dbl> 50, 47, 46, NA, NA, NA, NA, 36, 53, 67, 44, 62, NA, NA, NA, 4...
$ SEX_REF    <dbl> 1, 1, 2, 1, 2, 1, 2, 1, 1, 2, 2, 2, 2, 2, 2, 2, 2, 2, 1, 2, 1...
$ SEX2       <dbl> 2, 2, 1, NA, NA, NA, NA, 2, 2, 1, 1, 1, NA, NA, NA, 1, NA, 1,...
$ REF_RACE   <dbl> 1, 4, 2, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 2, 1, 1, 1, 1, 1...
$ RACE2      <dbl> 1, 4, 1, NA, NA, NA, NA, 1, 1, 1, 1, 1, NA, NA, NA, 2, NA, 1,...
$ HISP_REF   <dbl> 2, 2, 2, 2, 2, 1, 1, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 1, 1...
$ HISP2      <dbl> 2, 2, 1, NA, NA, NA, NA, 2, 2, 2, 2, 2, NA, NA, NA, 2, NA, 2,...
$ FAM_TYPE   <dbl> 3, 4, 1, 8, 9, 9, 8, 3, 1, 1, 3, 1, 8, 9, 8, 5, 9, 4, 8, 3, 2...
$ MARITAL1   <dbl> 1, 1, 1, 5, 3, 3, 2, 1, 1, 1, 1, 1, 2, 3, 5, 1, 3, 1, 3, 1, 1...
$ REGION     <dbl> 4, 4, 3, 4, 4, 3, 4, 1, 3, 2, 1, 4, 1, 3, 3, 3, 2, 1, 2, 4, 3...
$ SMSASTAT   <dbl> 1, 1, 1, 1, 1, 1, 1, 1, 2, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1...
```

Wrangling CE Data

Want to better understand a family's income and expenditures

```
1 ce <- ce_raw %>%
2   select(NEWID, PRINEARN, FINCBTAX,
3         BLS_URBN, HIGH_EDU, TOTEXPCQ, IRAX)
4 dim(ce)

[1] 6301      7
```

Variables:

- **NEWID**: ID for the household
- **PRINEARN**: ID for which member of the household is the principal earner
- **FINCBTAX**: Final income before taxes for the year
- **BLS_URBN**: 1 = urban, 2 = rural
- **HIGH_EDU**: Highest education in the household. 00 = Never attended, 10 = Grades 1-8, 11 = Grades 9-12, no degree, 12 = High school graduate, 13 = Some college, no degree, 14 = Associates degree, 15 = Bachelor's degree, 16 = Masters, Professional/doctorate degree
- **TOTEXPCQ** = Total household expenditures for the current quarter
- **IRAX** = Total in retirement funds

Wrangling CE Data

```
1 ce <- ce %>%
2   mutate(YEARLY_EXP = TOTEXPCQ*4)
3 ce
```

A tibble: 6,301 × 8

	NEWID	PRINEARN	FINCBTAX	BLS_URBN	HIGH_EDU	TOTEXPCQ	IRAX	YEARLY_EXP
	<chr>	<chr>	<dbl>	<dbl>	<chr>	<dbl>	<dbl>	<dbl>
1	03324174	01	116920	1	16	0	1000000	0
2	03324204	01	200	1	15	0	10000	0
3	03324214	01	117000	1	16	0	0	0
4	03324244	01	0	1	15	0	NA	0
5	03324274	02	2000	1	14	0	NA	0
6	03324284	01	942	1	11	0	0	0
7	03324294	01	0	1	10	0	0	0
8	03324304	01	91000	1	15	0	15000	0
9	03324324	02	95000	2	15	0	NA	0
10	03324334	01	40037	1	14	0	477000	0

i 6,291 more rows

Logical Operators

```
1 ce_sub <- ce %>%
2   filter(YEARLY_EXP > 0, BLS_URBN == 1, HIGH_EDU != "00")
3 ce_sub
```

A tibble: 3,950 × 8

	NEWID	PRINEARN	FINCBTAX	BLS_URBN	HIGH_EDU	TOTEXPCQ	IRAX	YEARLY_EXP
	<chr>	<chr>	<dbl>	<dbl>	<chr>	<dbl>	<dbl>	<dbl>
1	03335204	01	37000	1	14	2492.	0	9968.
2	03335214	01	103000	1	16	6128.	NA	24513.
3	03335224	01	14686	1	13	1072.	NA	4287.
4	03335244	02	33396	1	12	1630	0	6520
5	03335264	01	0	1	13	3213.	NA	12853.
6	03335274	01	0	1	15	4674.	0	18694.
7	03335294	01	745136	1	16	8693.	280000	34773.
8	03335304	01	36000	1	16	3733.	NA	14933.
9	03335314	02	45000	1	15	3627.	3000	14509
10	03335334	01	20862	1	13	802.	0	3209.

i 3,940 more rows

Logical Operators

```
1 ce_sub <- ce %>%
2   filter(YEARLY_EXP > 0, (BLS_URBN == 1 | HIGH_EDU != "00"))
3 ce_sub
```

```
# A tibble: 4,178 × 8
  NEWID      PRINEARN FINCBTAX BLS_URBN HIGH_EDU TOTEXPCQ   IRAX YEARLY_EXP
  <chr>      <chr>      <dbl>   <dbl> <chr>      <dbl> <dbl>      <dbl>
1 03335204 01          37000       1 14        2492.     0        9968.
2 03335214 01         103000       1 16        6128.    NA       24513.
3 03335224 01          14686       1 13        1072.    NA        4287.
4 03335244 02          33396       1 12         1630     0         6520
5 03335264 01           0       1 13        3213.    NA       12853.
6 03335274 01           0       1 15        4674.     0       18694.
7 03335294 01        745136       1 16        8693. 280000     34773.
8 03335304 01          36000       1 16        3733.    NA       14933.
9 03335314 02          45000       1 15        3627.   3000     14509
10 03335334 01          20862       1 13         802.     0         3209.
# i 4,168 more rows
```


case_when: Recoding Variables

```
1 count(ce, BLS_URBN)
```

```
# A tibble: 2 × 2
```

```
  BLS_URBN      n  
    <dbl> <int>
```

```
1         1  5952
```

```
2         2   349
```

```
1 ce <- ce %>%  
2   mutate(BLS_URBN = case_when(  
3     BLS_URBN == 1 ~ "Urban",  
4     BLS_URBN == 2 ~ "Rural"  
5   ))  
6 count(ce, BLS_URBN)
```

```
# A tibble: 2 × 2
```

```
  BLS_URBN      n  
    <chr>    <int>
```

```
1 Rural      349
```

```
2 Urban     5952
```

case_when: Creating Variables

```
1 count(ce, HIGH_EDU)
```

```
# A tibble: 8 × 2
```

	HIGH_EDU	n
	<chr>	<int>
1	00	8
2	10	110
3	11	302
4	12	1272
5	13	1297
6	14	714
7	15	1528
8	16	1070

```
1 ce <- ce %>%
2   mutate(HIGH_EDU = as.numeric(HIGH_EDU))
3 count(ce, HIGH_EDU)
```

```
# A tibble: 8 × 2
```

	HIGH_EDU	n
	<dbl>	<int>
1	0	8
2	10	110
3	11	302
4	12	1272
5	13	1297
6	14	714
7	15	1528
8	16	1070

```
1 ce <- ce %>%
2   mutate(HIGH_EDU2 = case_when(
3     is.na(HIGH_EDU) ~ NA,
4     HIGH_EDU <= 11 ~ "Less than high school degree",
5     between(HIGH_EDU, 12, 13) ~ "High school degree",
6     HIGH_EDU >= 14 ~ "College degree"
7   ))
8 count(ce, HIGH_EDU2)
```

```
# A tibble: 3 × 2
```

	HIGH_EDU2	n
	<chr>	<int>
1	College degree	3312
2	High school degree	2569
3	Less than high school degree	420

Variable Names

Sometimes datasets come with terrible variable names.

```
1 ce <- ce %>%
2   rename(INCOME = FINCBTAX)
3 ce
```

```
# A tibble: 6,301 × 9
  NEWID PRINEARN INCOME BLS_URBN HIGH_EDU TOTEXPCQ IRAX YEARLY_EXP HIGH_EDU2
  <chr> <chr>    <dbl> <chr>    <dbl>    <dbl>    <dbl>    <dbl>    <chr>
1 0332... 01      116920 Urban    16        0 1000000      0 College ...
2 0332... 01         200 Urban    15        0  10000      0 College ...
3 0332... 01      117000 Urban    16        0      0      0 College ...
4 0332... 01         0 Urban    15        0    NA      0 College ...
5 0332... 02       2000 Urban    14        0    NA      0 College ...
6 0332... 01       942 Urban    11        0      0      0 Less tha...
7 0332... 01         0 Urban    10        0      0      0 Less tha...
8 0332... 01      91000 Urban    15        0  15000      0 College ...
9 0332... 02      95000 Rural    15        0    NA      0 College ...
10 0332... 01     40037 Urban    14        0 477000      0 College ...
# i 6,291 more rows
```

Handling Missing Data

Want to compute mean income and mean retirement funds by location.

```
1 ce %>%
2   group_by(BLS_URBN) %>%
3   summarize(mean_INCOME = mean(INCOME),
4             mean_IRAX = mean(IRAX),
5             households = n())
```

A tibble: 2 × 4

	BLS_URBN <chr>	mean_INCOME <dbl>	mean_IRAX <dbl>	households <int>
1	Rural	40440.	NA	349
2	Urban	63772.	NA	5952

```
1 ce_aggressive <- ce_raw %>%
2   na.omit()
3 ce_aggressive
```

A tibble: 0 × 51

i 51 variables: NEWID <chr>, PRINEARN <chr>, FINLWT21
<dbl>, FINCBTAX <dbl>,
BLS_URBN <dbl>, POPSIZE <dbl>, EDUC_REF <chr>, EDUCA2
<dbl>, AGE_REF <dbl>,
AGE2 <dbl>, SEX_REF <dbl>, SEX2 <dbl>, REF_RACE <dbl>,
RACE2 <dbl>,
HISP_REF <dbl>, HISP2 <dbl>, FAM_TYPE <dbl>, MARITAL1
<dbl>, REGION <dbl>,
SMSASTAT <dbl>, HIGH_EDU <chr>, EHOUSNGC <dbl>,
TOTEXPCQ <dbl>,
FOODCQ <dbl>, TRANSCQ <dbl>, HEALTHCQ <dbl>, ENTERTCQ
<dbl>, EDUCACQ <dbl>,
TOBACCCQ <dbl>, STUDFINX <dbl>, IRAX <dbl>, CUTENURE
<dbl>, ...

Handling Missing Data

```
1 ce_moderate <- ce %>%
2   drop_na(IRAX, INCOME, BLS_URBN) %>%
3   group_by(BLS_URBN) %>%
4   summarize(mean_INCOME = mean(INCOME),
5             mean_IRAX = mean(IRAX),
6             households = n())
7
8 ce_moderate
```

```
# A tibble: 2 × 4
  BLS_URBN mean_INCOME mean_IRAX households
  <chr>      <dbl>      <dbl>      <int>
1 Rural      38651.      37008.         63
2 Urban      58987.      94512.        991
```

```
1 ce_light <- ce %>%
2   group_by(BLS_URBN) %>%
3   summarize(mean_INCOME = mean(INCOME, na.rm = TRUE),
4             mean_IRAX = mean(IRAX, na.rm = TRUE),
5             households = n())
6
7 ce_light
```

```
# A tibble: 2 × 4
  BLS_URBN mean_INCOME mean_IRAX households
  <chr>      <dbl>      <dbl>      <int>
1 Rural      40440.      37008.         349
2 Urban      63772.      94512.        5952
```

Multiple Groupings

```
1 ce %>%
2   group_by(BLS_URBN, HIGH_EDU2) %>%
3   summarize(mean_INCOME = mean(INCOME, na.rm = TRUE),
4             mean_IRAX = mean(IRAX, na.rm = TRUE),
5             households = n()) %>%
6   arrange(mean_IRAX)
```

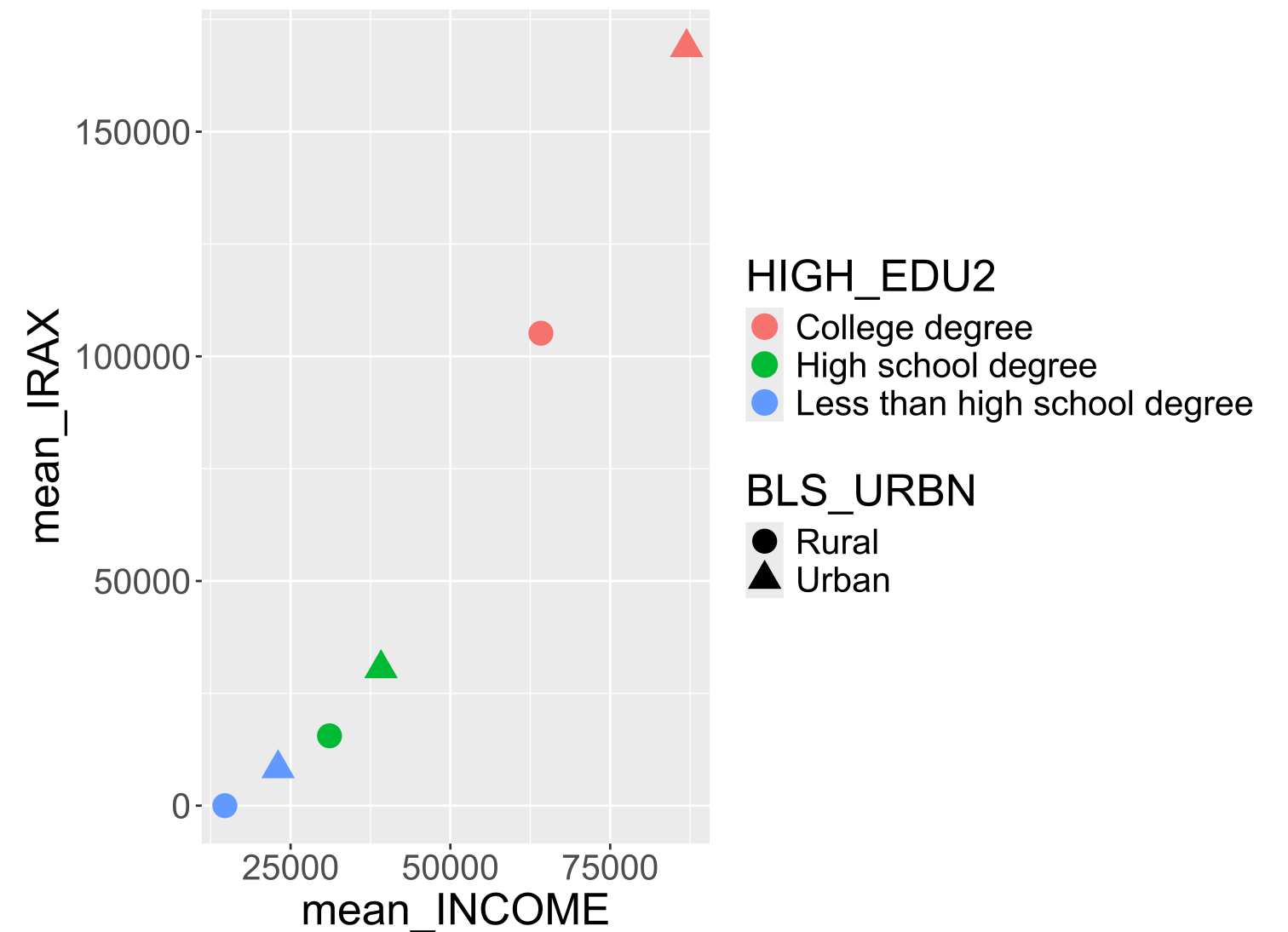
A tibble: 6 × 5

Groups: BLS_URBN [2]

	BLS_URBN	HIGH_EDU2	mean_INCOME	mean_IRAX	households
	<chr>	<chr>	<dbl>	<dbl>	<int>
1	Rural	Less than high school degree	14715.	0	39
2	Urban	Less than high school degree	23046.	8270.	381
3	Rural	High school degree	31087.	15543.	192
4	Urban	High school degree	39147.	30533.	2377
5	Rural	College degree	64161.	105148.	118
6	Urban	College degree	86957.	168767.	3194

Piping into ggplot2

```
1 ce %>%
2   group_by(BLS_URBN, HIGH_EDU2) %>%
3   summarize(mean_INCOME = mean(INCOME, na.rm = TRUE),
4             mean_IRAX = mean(IRAX, na.rm = TRUE),
5             households = n()) %>%
6   ggplot(mapping = aes(x = mean_INCOME,
7                       y = mean_IRAX,
8                       shape = BLS_URBN,
9                       color = HIGH_EDU2)) +
10  geom_point(size = 5)
```



Data Joins

- Often in the data analysis workflow, we have more than one data source, which means more than one dataframe, and we want to combine these dataframes.
- Need principled way to combine.
 - Need a **key** that links two dataframes together.
- These multiple dataframes are called **relational data**.

CE Data

- **Household** survey but data are also collected on **individuals**
 - **fml**i: household data
 - **mem**i: household member-level data

```
1 fml_i <- read_csv("data/fml_i.csv",  
2                   na = c("NA", ".")) %>%  
3   select(NEWID, PRINEARN, FINCBTAX,  
4          BLS_URBN, HIGH_EDU)  
5 mem_i <- read_csv("data/mem_i.csv",  
6                   na = c("NA", ".")) %>%  
7   select(NEWID, MEMBNO, AGE, SEX, EARNTYPE)
```

- Want to add variables on the **principal earner** from the member data frame to the household data frame

CE Data

Key variable(s)?

```
1 fml_i
```

```
# A tibble: 6,301 × 5
  NEWID    PRINEARN FINCBTAX BLS_URBN HIGH_EDU
  <chr>    <chr>      <dbl>   <dbl> <chr>
1 03324174 01        116920     1 16
2 03324204 01         200     1 15
3 03324214 01       117000     1 16
4 03324244 01          0     1 15
5 03324274 02        2000     1 14
6 03324284 01         942     1 11
7 03324294 01          0     1 10
8 03324304 01       91000     1 15
9 03324324 02       95000     2 15
10 03324334 01       40037     1 14
# i 6,291 more rows
```

```
1 mem_i
```

```
# A tibble: 15,412 × 5
  NEWID    MEMBNO    AGE    SEX EARNTYPE
  <chr>    <dbl> <dbl> <dbl>   <dbl>
1 03552611     1    58     2     2
2 03552641     1    54     1     1
3 03552641     2    49     2    NA
4 03552651     1    39     2    NA
5 03552651     2    10     2    NA
6 03552651     3    32     1    NA
7 03552651     4     7     1    NA
8 03552651     5     9     1    NA
9 03552681     1    38     1     3
10 03552681     2    34     2    NA
# i 15,402 more rows
```

CE Data

- Key variables?
 - Problem with class?

```
1 class(fmli$NEWID)
```

```
[1] "character"
```

```
1 class(memi$NEWID)
```

```
[1] "character"
```

```
1 class(fmli$PRINEARN)
```

```
[1] "character"
```

```
1 class(memi$MEMBNO)
```

```
[1] "numeric"
```

CE Data

- Key variables?
 - Problem with class?

```
1 fml_i <- mutate(fml_i, PRINEARN = as.integer(PRINEARN))  
2 class(fml_i$PRINEARN)
```

```
[1] "integer"
```

```
1 class(memi$MEMBNO)
```

```
[1] "numeric"
```

CE Data

- Want to add columns of `memi` to `fml` that correspond to the principal earner's memi data
 - What type of join is that?

The World of Joins

- **Mutating joins**: Add new variables to one dataset from matching observations in another.
 - `left_join()` (and `right_join()`)
 - `inner_join()`
 - `full_join()`
- There are also *filtering* joins but we won't cover those today.

Example Dataframes

Here I created the data frames by hand.

```
1 # Note: These are fake DATA 100 students
2 students <- data.frame(name = c("Alice", "Josh", "Megan", "Eric", "Jean"),
3                         res_hall = c("South Campus Apts", "Smith", "Off-campus", "Smith", "Trax"),
4                         caf = c("like", "hate", "hate", "neutral", "love"))
5
6 halls <- data.frame(name = c("Kress", "Smith", "South Campus Apts", "Trax"),
7                     steps = c(250, 450, 1400, 400))
```

Example Dataframes

1 students				
	name	res_hall	caf	
1	Alice	South Campus Apts	like	
2	Josh	Smith	hate	
3	Megan	Off-campus	hate	
4	Eric	Smith	neutral	
5	Jean	Trax	love	
1 halls				

	name	steps
1	Kress	250
2	Smith	450
3	South Campus Apts	1400
4	Trax	400

left_join()

```
1 students_new <- left_join(students, halls)
```

Joining with `by = join\_by(name)`

```
1 students_new
```

	name	res_hall	caf	steps
1	Alice	South Campus	Apts like	NA
2	Josh	Smith	hate	NA
3	Megan	Off-campus	hate	NA
4	Eric	Smith	neutral	NA
5	Jean	Trax	love	NA

left_join()

```
1 students_new <- left_join(students, halls, join_by("res_hall" == "name"))
2 students_new
```

	name	res_hall	caf	steps
1	Alice	South Campus	Apts like	1400
2	Josh	Smith	hate	450
3	Megan	Off-campus	hate	NA
4	Eric	Smith	neutral	450
5	Jean	Trax	love	400

inner_join()

```
1 students_new <- inner_join(students, halls, join_by("res_hall" == "name"))
2 students_new
```

	name	res_hall	caf	steps
1	Alice	South Campus	Apts like	1400
2	Josh	Smith	hate	450
3	Eric	Smith	neutral	450
4	Jean	Trax	love	400

full_join()

```
1 students_new <- full_join(students, halls, join_by("res_hall" == "name"))
2 students_new
```

	name	res_hall	caf	steps
1	Alice	South Campus	Apts like	1400
2	Josh	Smith	hate	450
3	Megan	Off-campus	hate	NA
4	Eric	Smith	neutral	450
5	Jean	Trax	love	400
6	<NA>	Kress	<NA>	250

Back to our Real-World Example

- What kind of join do we want for the Consumer Expenditure data?
 - Want to add columns of **memi** to **fml**i that correspond to the principal earner's memi data
- Also going to create smaller data frames for us to play with:

```
1 fml_i_small <- filter(fml_i, NEWID %in% c("03530051",
2                                           "03327224",
3                                           "03324324",
4                                           "03324244"))
5 fml_i_small
```

A tibble: 4 × 5

	NEWID	PRINEARN	FINCBTAX	BLS_URBN	HIGH_EDU
	<chr>	<int>	<dbl>	<dbl>	<chr>
1	03324244	1	0	1	15
2	03324324	2	95000	2	15
3	03327224	1	0	1	14
4	03530051	3	70000	1	11

```
1 memi_small <- filter(memi, NEWID %in% c("03530051",
2                                           "03327224",
3                                           "03324324",
4                                           "03324244"))
5 memi_small
```

A tibble: 10 × 5

	NEWID	MEMBNO	AGE	SEX	EARNTYPE
	<chr>	<dbl>	<dbl>	<dbl>	<dbl>
1	03324244	1	37	1	1
2	03324324	1	51	1	1
3	03324324	2	53	2	1
4	03327224	1	28	2	3
5	03327224	2	32	1	2
6	03327224	3	1	2	NA
7	03530051	1	43	1	NA
8	03530051	2	16	1	NA
9	03530051	3	44	1	3
10	03530051	4	5	2	NA

Look at the Possible Joins

```
1 left_join(fmli_small, memi_small)
```

Joining with `by = join\_by(NEWID)`

A tibble: 10 × 9

	NEWID	PRINEARN	FINCBTAX	BLS_URBN	HIGH_EDU	MEMBNO	AGE	SEX	EARNTYPE
	<chr>	<int>	<dbl>	<dbl>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>
1	03324244	1	0	1	15	1	37	1	1
2	03324324	2	95000	2	15	1	51	1	1
3	03324324	2	95000	2	15	2	53	2	1
4	03327224	1	0	1	14	1	28	2	3
5	03327224	1	0	1	14	2	32	1	2
6	03327224	1	0	1	14	3	1	2	NA
7	03530051	3	70000	1	11	1	43	1	NA
8	03530051	3	70000	1	11	2	16	1	NA
9	03530051	3	70000	1	11	3	44	1	3
10	03530051	3	70000	1	11	4	5	2	NA

Look at the Possible Joins

Be careful. This erroneous example made my R crash when I tried it on the full data frames.

```
1 left_join(fmli_small, memi_small, join_by("PRINEARN" == "MEMBNO"))
```

A tibble: 13 × 9

	NEWID.x	PRINEARN	FINCBTAX	BLS_URBN	HIGH_EDU	NEWID.y	AGE	SEX	EARNTYPE
	<chr>	<dbl>	<dbl>	<dbl>	<chr>	<chr>	<dbl>	<dbl>	<dbl>
1	03324244	1	0	1	15	03324244	37	1	1
2	03324244	1	0	1	15	03324324	51	1	1
3	03324244	1	0	1	15	03327224	28	2	3
4	03324244	1	0	1	15	03530051	43	1	NA
5	03324324	2	95000	2	15	03324324	53	2	1
6	03324324	2	95000	2	15	03327224	32	1	2
7	03324324	2	95000	2	15	03530051	16	1	NA
8	03327224	1	0	1	14	03324244	37	1	1
9	03327224	1	0	1	14	03324324	51	1	1
10	03327224	1	0	1	14	03327224	28	2	3
11	03327224	1	0	1	14	03530051	43	1	NA
12	03530051	3	70000	1	11	03327224	1	2	NA
13	03530051	3	70000	1	11	03530051	44	1	3

```
1 count(fmli_small, PRINEARN)
```

A tibble: 3 × 2

	PRINEARN	n
	<int>	<int>
1	1	2
2	2	1
3	3	1

```
1 count(memi_small, MEMBNO)
```

A tibble: 4 × 2

	MEMBNO	n
	<dbl>	<int>
1	1	4
2	2	3
3	3	2
4	4	1

Look at the Possible Joins

```
1 left_join(fmli_small, memi_small, join_by("NEWID" == "NEWID", "PRINEARN" == "MEMBNO"))
```

A tibble: 4 × 8

	NEWID	PRINEARN	FINCBTAX	BLS_URBN	HIGH_EDU	AGE	SEX	EARNTYPE
	<chr>	<dbl>	<dbl>	<dbl>	<chr>	<dbl>	<dbl>	<dbl>
1	03324244	1	0	1	15	37	1	1
2	03324324	2	95000	2	15	53	2	1
3	03327224	1	0	1	14	28	2	3
4	03530051	3	70000	1	11	44	1	3

Look at the Possible Joins

```
1 inner_join(fmli_small, memi_small, join_by("NEWID" == "NEWID", "PRINEARN" == "MEMBNO"))
```

```
# A tibble: 4 × 8
```

	NEWID <chr>	PRINEARN <dbl>	FINCBTAX <dbl>	BLS_URBN <dbl>	HIGH_EDU <chr>	AGE <dbl>	SEX <dbl>	EARNTYPE <dbl>
1	03324244	1	0	1	15	37	1	1
2	03324324	2	95000	2	15	53	2	1
3	03327224	1	0	1	14	28	2	3
4	03530051	3	70000	1	11	44	1	3

Why does this give us the same answer as `left_join` for this situation?

Look at the Possible Joins

```
1 full_join(fmli_small, memi_small, join_by("NEWID" == "NEWID", "PRINEARN" == "MEMBNO"))
```

A tibble: 10 × 8

	NEWID	PRINEARN	FINCBTAX	BLS_URBN	HIGH_EDU	AGE	SEX	EARNTYPE
	<chr>	<dbl>	<dbl>	<dbl>	<chr>	<dbl>	<dbl>	<dbl>
1	03324244	1	0	1	15	37	1	1
2	03324324	2	95000	2	15	53	2	1
3	03327224	1	0	1	14	28	2	3
4	03530051	3	70000	1	11	44	1	3
5	03324324	1	NA	NA	<NA>	51	1	1
6	03327224	2	NA	NA	<NA>	32	1	2
7	03327224	3	NA	NA	<NA>	1	2	NA
8	03530051	1	NA	NA	<NA>	43	1	NA
9	03530051	2	NA	NA	<NA>	16	1	NA
10	03530051	4	NA	NA	<NA>	5	2	NA

Joining Tips

```
1 fml_i <- left_join(fml_i, mem_i, join_by("NEWID" == "NEWID", "PRINEARN" == "MEMBNO"))
```

- FIRST: conceptualize for yourself what you think you want the final dataset to look like!
- Check initial dimensions and final dimensions.
- Use variable names when joining even if they are the same.

Naming Wrangled Data

Should I name my new dataframe **ce** or **ce1**? In other words, should I write over my original dataframe or should I save a new dataframe?

- *My answer:*
 - Is your new dataframe structurally different? If so, give it a **new name**.
 - Are you removing values you will need for a future analysis within the same document? If so, give it a **new name**.
 - Are you just adding to or cleaning the data? If so, then **write over** the original.

Live Coding

Sage Advice from ModernDive

“Crucial: Unless you are very confident in what you are doing, it is worthwhile not starting to code right away. Rather, first sketch out on paper all the necessary data wrangling steps not using exact code, but rather high-level pseudocode that is informal yet detailed enough to articulate what you are doing. This way you won’t confuse what you are trying to do (the algorithm) with how you are going to do it (writing dplyr code).”